

Univariate Data Visualization

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1 Univariate Data Visualizations

```
library(ggplot2)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(papaja)
```

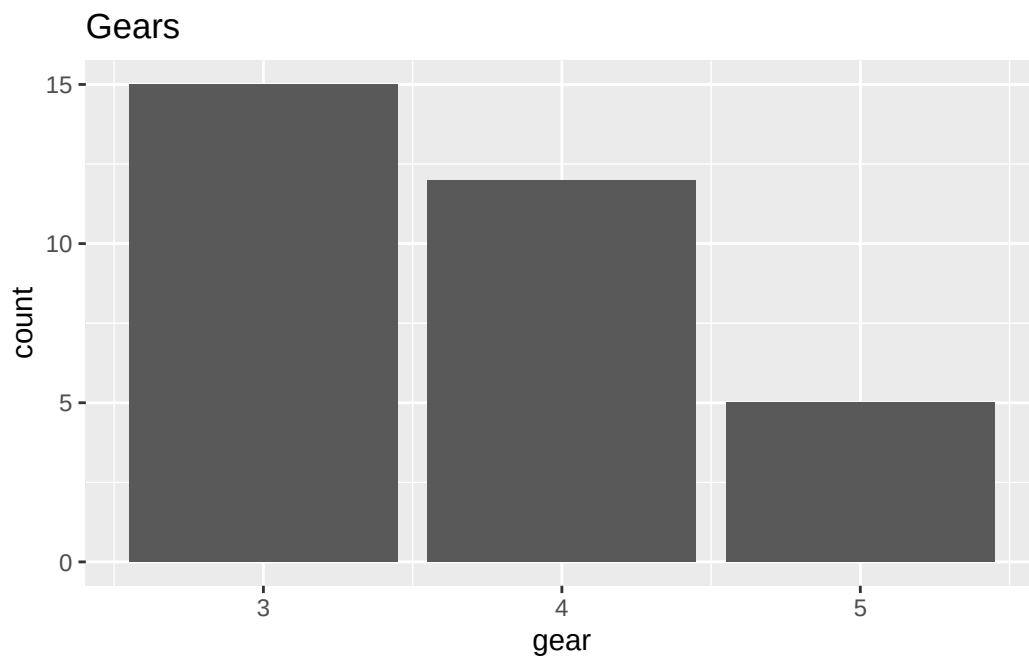
Loading required package: tinylab

```
# data(mtcars)
df <- mtcars
names(df)
```

```
[1] "mpg"  "cyl"  "disp" "hp"    "drat" "wt"    "qsec" "vs"    "am"    "gear"
[11] "carb"
```

1.1 Figure 1 Bar Graph

```
# univariate graph (categorical variable)
df %>%
  ggplot(aes(x=gear))+
  geom_bar()+
  ggtitle("Gears")
```

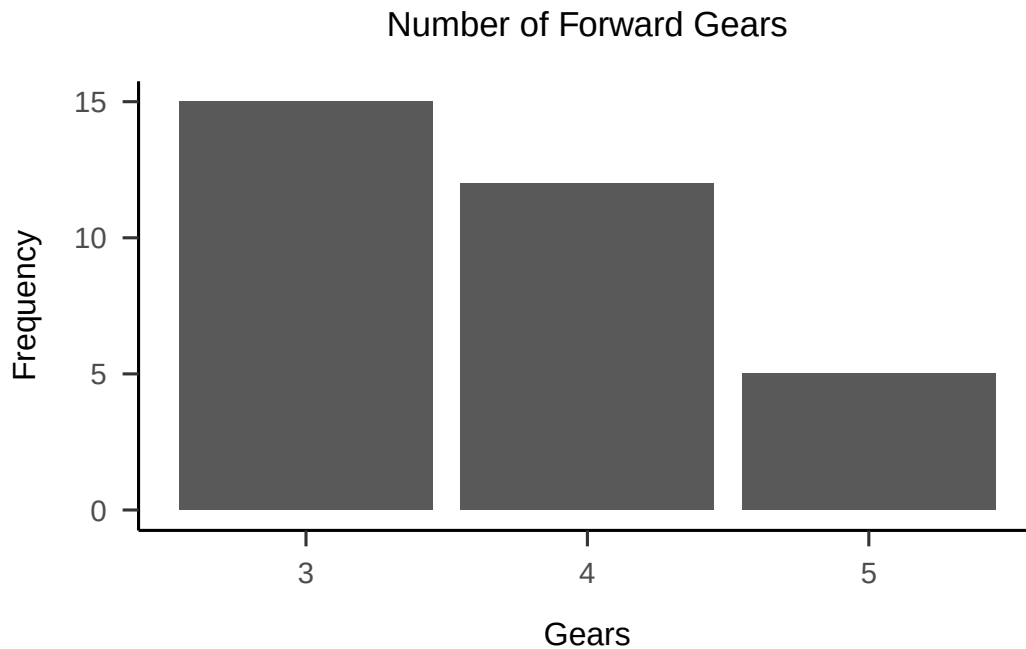


```
# univariate graph (categorical variable)
df %>%
  ggplot(aes(x=gear)) +
  geom_bar() +
  ggtitle("Number of Forward Gears") +
```

```

xlab("Gears") +
ylab("Frequency") +
theme_apapa() # from papaja package

```



1.2 Figure 2 Histogram

```
names(mtcars)
```

```

[1] "mpg" "cyl" "disp" "hp" "drat" "wt" "qsec" "vs" "am" "gear"
[11] "carb"

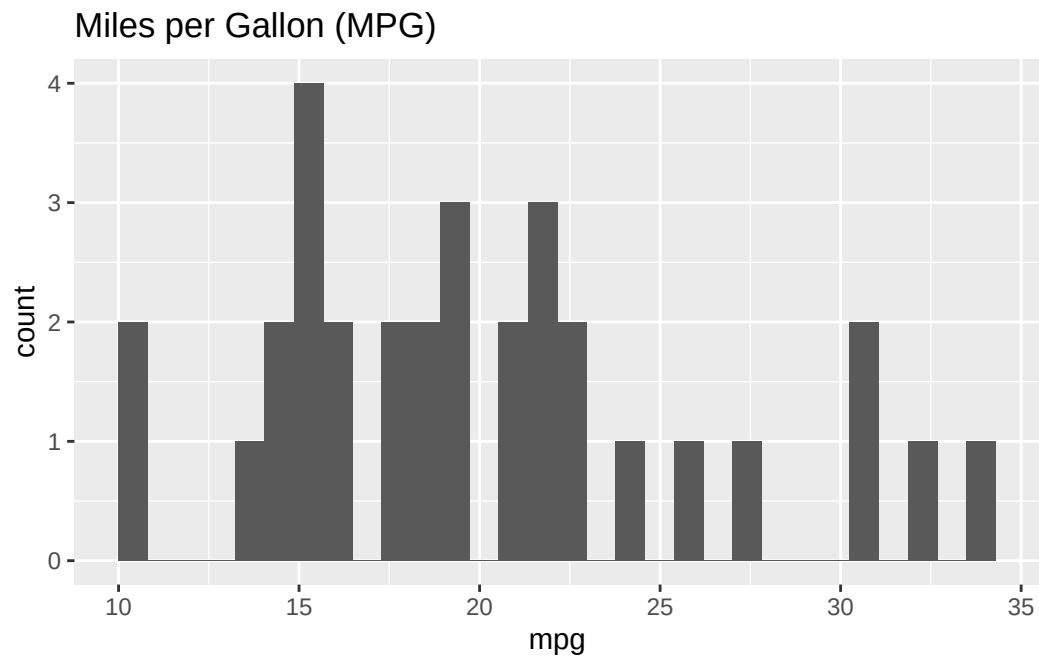
```

```

# univariate graph (categorical variable)
df %>%
  ggplot(aes(x=mpg)) +
  geom_histogram() +
  ggtitle("Miles per Gallon (MPG)")

```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



```
# univariate graph (categorical variable)
df %>%
  filter(!is.na(mpg)) %>%
  ggplot(aes(x=mpg)) +
  geom_histogram(binwidth = 5, fill = "lightblue", color = "black") +
  ggtitle("Miles per Gallon (MPG)") +
  theme_apapa()
```

